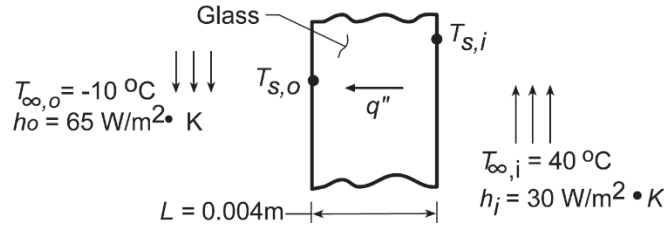


Heat Transfer (EMCH 354), Spring 2023

Homework #3 Solution

Problem #1



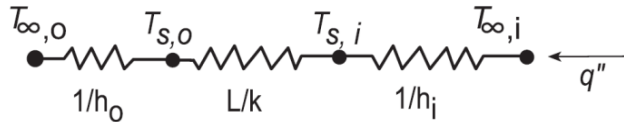
a) Inner and outer surface temperature?

$$q'' = \frac{T_{S,o} - T_{\infty,o}}{\frac{1}{h_o}} = \frac{T_{S,i} - T_{S,o}}{\frac{L}{k}} = \frac{T_{\infty,i} - T_{S,i}}{\frac{1}{h_i}} = \frac{T_{\infty,i} - T_{\infty,o}}{\frac{1}{h_o} + \frac{L}{k} + \frac{1}{h_i}}$$

$$q'' = \frac{T_{\infty,i} - T_{\infty,o}}{\frac{1}{h_o} + \frac{L}{k} + \frac{1}{h_i}} = \frac{40 - (-10)}{\frac{1}{60} + \frac{0.004}{1.4} + \frac{1}{30}} = \frac{50}{0.01666 + 0.00286 + 0.0333} = 945.95 \text{ W/m}^2$$

$$q'' = \frac{T_{S,o} - T_{\infty,o}}{\frac{1}{h_o}} \Rightarrow 945.95 = \frac{T_{S,o} - (-10)}{\frac{1}{60}} \Rightarrow T_{S,o} = 5.766 \text{ } ^\circ\text{C}$$

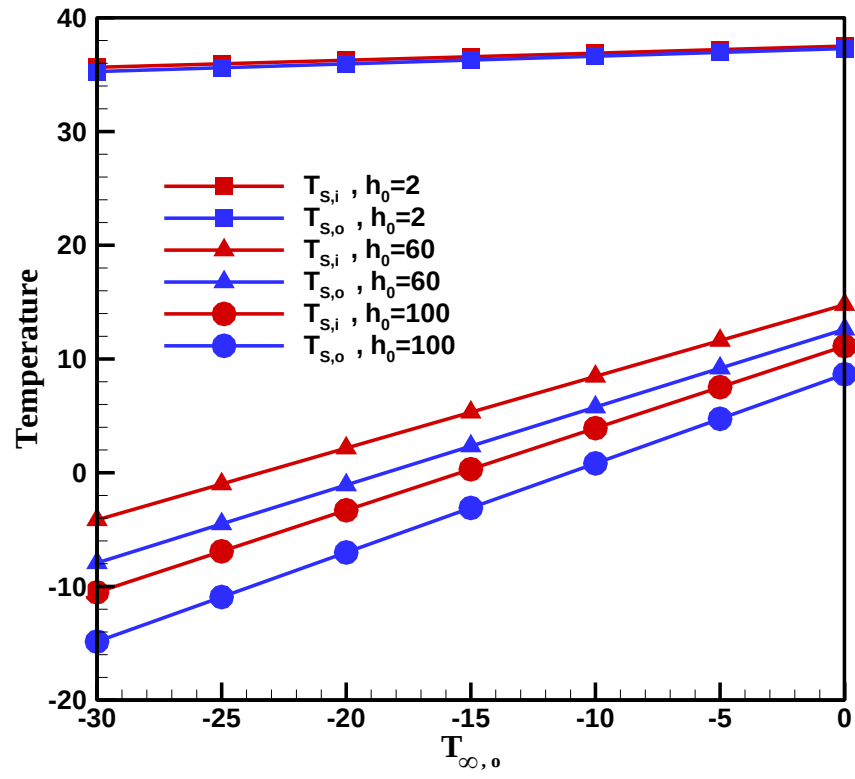
$$q'' = \frac{T_{\infty,i} - T_{S,i}}{\frac{1}{h_i}} \Rightarrow 945.95 = \frac{40 - T_{S,i}}{\frac{1}{30}} \Rightarrow T_{S,i} = 8.468 \text{ } ^\circ\text{C}$$



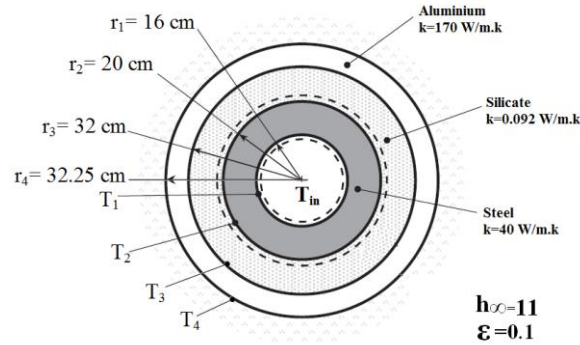
b) Plot inner and outer surface temperatures?

T_{∞}	$h_{\infty,o}=2$			$h_{\infty,o}=60$			$h_{\infty,o}=100$		
	q''	$T_{S,i}$	$T_{S,o}$	q''	$T_{S,i}$	$T_{S,o}$	q''	$T_{S,i}$	$T_{S,o}$
-30	130.55	35.65	35.28	1324.32	-4.14	-7.93	1515.46	-10.52	-14.85
-25	121.23	35.96	35.61	1229.73	-0.99	-4.50	1407.22	-6.91	-10.93
-20	111.90	36.27	35.95	1135.14	2.16	-1.08	1298.97	-3.30	-7.01
-15	102.58	36.58	36.29	1040.54	5.32	2.34	1190.72	0.31	-3.09
-10	93.25	36.89	36.63	945.95	8.47	5.77	1082.47	3.92	0.82
-5	83.93	37.20	36.96	851.35	11.62	9.19	974.23	7.53	4.74

0	74.60	37.51	37.30	756.76	14.77	12.61	865.98	11.13	8.66
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Problem # 2



$$R = R_1 + R_2 + R_3 + \left(\frac{1}{R_3} + \frac{1}{R_4} \right)^{-1} \Rightarrow$$

$$R = \frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi k_1 L} + \frac{\ln\left(\frac{r_3}{r_2}\right)}{2\pi k_2 L} + \frac{\ln\left(\frac{r_4}{r_3}\right)}{2\pi k_3 L} + \left(\frac{1}{\frac{1}{h_{\infty} A}} + \frac{1}{\frac{1}{h_R A}} \right)^{-1} \Rightarrow$$

$$R = \frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi k_1 L} + \frac{\ln\left(\frac{r_3}{r_2}\right)}{2\pi k_2 L} + \frac{\ln\left(\frac{r_4}{r_3}\right)}{2\pi k_3 L} + \left(\frac{1}{\frac{1}{h_{\infty} 2\pi r_4 L}} + \frac{1}{\frac{1}{h_R 2\pi r_4 L}} \right)^{-1} \Rightarrow$$

$$R = \frac{1}{2\pi L} \left(\frac{\ln\left(\frac{r_2}{r_1}\right)}{k_1} + \frac{\ln\left(\frac{r_3}{r_2}\right)}{k_2} + \frac{\ln\left(\frac{r_4}{r_3}\right)}{k_3} + (h_{\infty} r_4 + h_R r_4)^{-1} \right) \Rightarrow$$

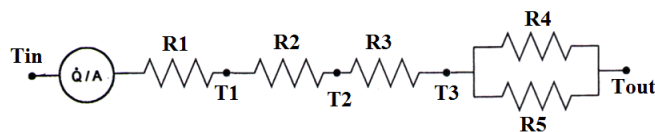
$$h_R = 4\epsilon\sigma T_m^3 = 4 \times 0.1 \times 5.67 \times 10^{-8} \times (26 + 273.15)^3 = 0.61 \text{ W/m}^2 \cdot k$$

$$R = \frac{1}{2\pi L} \left(\frac{\ln\left(\frac{20}{16}\right)}{40} + \frac{\ln\left(\frac{32}{20}\right)}{0.092} + \frac{\ln\left(\frac{32.25}{32}\right)}{170} + (11 \times 0.3225 + 0.61 \times 0.3225)^{-1} \right) \Rightarrow$$

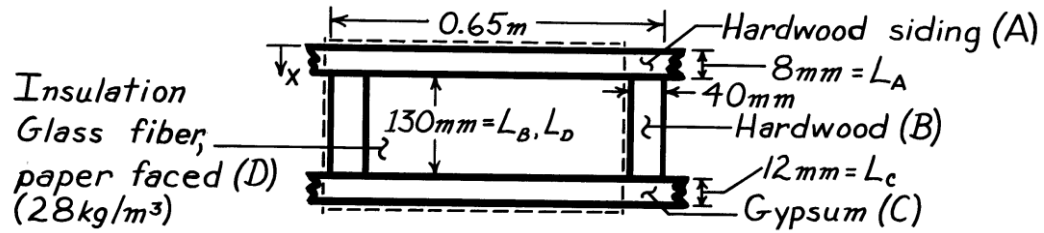
$$R = \frac{1}{2\pi L} (5.58 \times 10^{-3} + 5.11 + 4.58 \times 10^{-5} + 2.67 \times 10^{-1}) = \frac{5.38}{2\pi L} \Rightarrow$$

$$Q = \frac{\Delta T}{R} \Rightarrow Q = \frac{565 - 26}{\frac{5.38}{2\pi L}} \Rightarrow \frac{Q}{L} = \frac{565 - 26}{\frac{5.38}{2\pi}} = 629.49$$

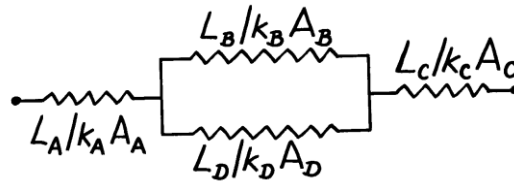
Please note: L could be set as 1 at the very beginning for simplicity.



Problem # 3



Using the isothermal surface assumption, the thermal circuit associated with a single unit (enclosed by dashed lines) of the wall is:



$$R_{eq} = R_A + \left(\frac{1}{R_B} + \frac{1}{R_D} \right)^{-1} + R_C$$

$$R_A = \frac{L_A}{k_A A_A} = \frac{0.008}{0.094 \times (0.65 \times 2.5)} = 0.0524 \text{ K/W}$$

$$R_B = \frac{L_B}{k_B A_B} = \frac{0.13}{0.16 \times (0.04 \times 2.5)} = 8.125 \text{ K/W}$$

$$R_D = \frac{L_D}{k_D A_D} = \frac{0.13}{0.038 \times (0.61 \times 2.5)} = 2.243 \text{ K/W}$$

$$R_C = \frac{L_C}{k_C A_C} = \frac{0.012}{0.17 \times (0.65 \times 2.5)} = 0.0434 \text{ K/W}$$

$$R_{eq} = R_A + \left(\frac{1}{R_B} + \frac{1}{R_D} \right)^{-1} + R_C = 0.0524 + \left(\frac{1}{8.125} + \frac{1}{2.243} \right)^{-1} + 0.0434$$

$$R_{eq} = 0.0524 + 1.758 + 0.0434 = 1.854 \text{ K/W}$$

With 10 such units in parallel, the total wall resistance is:

Problem # 4

$$P = 2\pi R = 2\pi \times 2 \times 10^{-3} \text{ m}$$

$$A = \pi R^2 = \pi \times (2 \times 10^{-3})^2 \text{ m}^2$$

$$\beta = \sqrt{\frac{h \times P}{k \times A}} = \sqrt{\frac{h \times 2\pi R}{k \times \pi R^2}} = \sqrt{\frac{h \times 2}{k \times R}} = \sqrt{\frac{100 \times 2}{180 \times 2 \times 10^{-3}}} = 23.57 \text{ m}^{-1}$$

$$\beta \cdot L = 23.57 \times 20 \times 10^{-3} = 0.471 \Rightarrow \beta \cdot L \ll 3 \Rightarrow \text{infinite fin}$$

$$Q_{fin} = \sqrt{k \times h \times P \times A} \times \tanh(m \cdot L) \theta_b$$

$$Q_{fin} = \sqrt{180 \times 100 \times 2\pi \times 2 \times 10^{-3} \times \pi \times (2 \times 10^{-3})^2} \times \tanh(0.471) (600 - 400) = 4.685 \text{ W}$$

$$\mathbf{Q_{fin} = 4.685 \text{ W}}$$

Problem # 5

Refer lecture and text book.